

Prompt 1: Split panel, full field of view

Photorealistic scientific figure showing two side-by-side 3D volume renderings of the same X-ray microscopy (XRM) tomography dataset of a lithium-ion battery cathode electrode. The overall field of view is approximately 150 x 150 micrometers.

Left panel (raw XRM rendering): grey-scale absorption contrast showing densely packed NMC cathode particles ranging from 5 to 20 micrometers in diameter. Cathode particles appear bright white to light grey due to high X-ray attenuation. Carbon black and binder matrix appears dark grey. Pore space appears black. Several particles show visible intragranular cracks as dark planar features in their interiors. Black background.

Right panel (segmented rendering of the identical volume): same particles and spatial arrangement as the left panel, with color segmentation applied. NMC cathode particles rendered in semi-transparent gold/amber. Carbon black and binder matrix in dark grey. Pore space in transparent blue. Cracked regions within particles highlighted in red, matching the crack locations visible in the left panel. Black background.

The two panels should show identical particle positions, sizes, and morphology, making clear they represent the same dataset before and after segmentation. Thin white vertical dividing line between panels. No labels, no scale bar, no annotations. Scientific publication quality, consistent with ZEISS Xradia Versa XRM data.

Prompt 2: High resolution zoomed-in, raw XRM rendering

Photorealistic 3D volume rendering of a single NMC cathode particle imaged at high resolution by X-ray microscopy (XRM), zoomed in to show intragranular crack morphology in detail. The particle is approximately 12 micrometers in diameter, rendered in bright grey-white absorption contrast typical of ZEISS Xradia Versa data. The particle surface shows slight roughness consistent with grain boundary faceting. The interior reveals a network of intragranular cracks running along grain boundaries, visible as dark planar features cutting through the particle interior in multiple orientations. Some cracks appear to connect, forming a network that extends from the particle interior to the surface. The surrounding carbon matrix is partially visible at the edges, appearing dark grey. The rendering uses cinematic 3D lighting with subtle shadows to enhance crack visibility. Black background. No labels, no scale bar, no annotations. Scientific publication quality.